HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Contributed to the design and development of mechanical components for water and wastewater treatment plants, aligning with project timelines and technical specifications.
- Assisted in site surveys of existing water treatment assets, gathering data to inform optioneering and feasibility studies for process upgrades.
- Prepared procurement specifications for mechanical equipment, ensuring compliance with project requirements and supplier quality standards.
- Collaborated within a multi-disciplined engineering team to produce mechanical design deliverables, including drawings and documentation across all project stages.
- Supported project delivery by liaising with subcontractors and suppliers, ensuring timely and accurate provision of mechanical equipment and services.
- Maintained Design Change Control logs and updated documentation to reflect modifications made during the design and implementation phases of projects.
- Developed Python scripts to automate routine data collection and reporting tasks for mechanical design projects, reducing manual effort by approximately 15%.

SKILLS

Engineering – Mechanical Design, Water Treatment, Wastewater Treatment, Pumping Systems, Site Surveys, Optioneering, Feasibility Studies, Procurement Specifications, Technical Requirements, Design Change Control, Health & Safety, CDM Regulations

Software & Tools – AutoCAD (inferred), SolidWorks (inferred), Microsoft Office Suite, Jira, Confluence

Programming & Scripting – Python, Shell Scripting

Databases & Data Handling – SQL, PostgreSQL, MySQL, Oracle

Cloud & Infrastructure – AWS, EC2, S3, Lambda, CloudWatch, Docker

Methodologies – Agile, Scrum

PROJECTS

Mechanical Design for Water Infrastructure Upgrade

- Designed key mechanical components for a regional water treatment facility upgrade, focusing on pumping systems and process interconnectivity.
- Created detailed construction drawings and technical specifications for procurement of new mechanical equipment, managing supplier engagement.
- Developed a Python script to parse and analyze sensor data from existing infrastructure, identifying potential bottlenecks and informing design improvements.

Wastewater Treatment Process Optimisation Study

- Contributed to a feasibility study for enhancing a wastewater treatment plant, including site surveys and analysis of existing mechanical systems.
- Modeled process flows and equipment layouts using CAD tools (inferred), simulating performance under various operational loads.
- Documented findings and recommendations for mechanical equipment upgrades, ensuring adherence to environmental and regulatory standards.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)